## Department of Computer Science and Engineering

### ISA EXAM I

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| Course : Object Oriented Programming | USN : |
| Course Code : 25ECSC204/24ECAC207 | Semester : IV |
| Date of Exam : 27/03/26 | Duration : 75 mins |

**Note: (i) Answer any two full questions. (ii) Each full question carries equal marks.**

| Q.No. | Questions | Marks |
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| 1.a | A university develops an exam result processing application in Java. The application is created on one machine but is later executed on computers having different operating systems (platform). The system wants to know how the same Java program can run everywhere without rewriting it and how it is done.<br>Q. Which feature of the Java language support the above scenario? JVM<br>A student designs a Library Management System to store and process book and member information. While the program is running, many objects are created dynamically, but the programmer does not explicitly remove unused objects from memory and it has to be done automatically. Also, situations such as invalid member ID, missing file, or null reference may arise during execution, and the program handles them effectively.<br>Q. Which feature of the Java language support the above scenario? | 4M |
| 1b | Write output of the following code snippet and comment on the output.<br>class Point2{ int xCoordinate;<br>int yCoordinate;<br>Point2(){<br>xCoordinate=0; yCoordinate=0;}<br>Point2(int x, int y){xCoordinate=x;<br>yCoordinate=y;}<br>Point2(Point2 obj)<br>{xCoordinate=obj.xCoordinate;<br>yCoordinate=obj.yCoordinate;}<br>void printPoint(){<br>System.out.println(xCoordinate+":"+yCoordinate); }<br>String toString(){<br>return xCoordinate+":"+yCoordinate;} }<br>class PointDemoDemo{<br>public static void main(String[] args){<br>Point2 p1=new Point2(10,5);<br>p1.printPoint();<br>Point2 p2=new Point2(p1);<br>p2.printPoint();<br>System.out.println(p2);} } | 6M |

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| 1c | Consider an Employee Management System that uses the class hierarchy: Employee(empId, name, salary, bonus), Manager(numberOfShares), and Salesman(numberOfSales). Assume that the initial bonus amount is zero for all employees. Define a method to pay bonus by considering the corresponding attribute of the employee while paying the bonus. Write a program to implement this class hierarchy and demonstrate run-time polymorphism. Also, explain whether run-time polymorphism improves the performance of the system. | 10M |
| 2a | Given the following set of classes, what type of relationship can be used? Draw the inheritance diagram. Is there any chance of making a class as abstract class in the given set of classes?<br>i. Employee and Manager<br>ii. CashCard, DebitCard , CreditCard and Customer<br>iii. BankAccount, SavingsAccount, CurrentAccount | 4M |
| 2b | Consider the following objects for Shopping Bill Payment System where a customer purchases three items and pays the total bill using a debit card. The system should calculate the total bill, and print the bill amount<br>i) Customer (custId, name, billAmt)<br>ii) Item (itemName, price)<br>iii) DebitCard (cardNo, balance)<br>Perform the following:<br>a) Identify the relationship between the objects<br>b) Identify the methods and add to the appropriate class<br>c) Draw the class diagram inclusive of all the classes and its attributes.<br>d) Currently, the system supports only three items. How would you modify the design to support any number of items dynamically? | 6M |
| 2c | Consider a laptop system, which is described as Laptop{lapTopId, brand, RAM, price}. Write a java code for the following:<br>i. Create the class using parameterized constructor and explain why it is preferred method in any system?<br>ii. Write an overloaded method to print laptops (one among should be method which take object array as parameter)<br>iii. Add a method to increase the price of all the laptops by 10%.<br>iv. Write a starter class to create objects and call methods. | 10M |
| 3a | In a library management system, every time a new Book record is entered, an object of the Book class is created. The system should keep track of the total number of Book objects created during execution. Write a program to count the number of books created, this count shall be initialized before any object of the class gets created. [Consider the book object only the library is to set the context.] | 4M |

ISA EXAM I

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| 3b | Write the output of the following code snippet and comment on the output.<br>class Car<br>{ private int fuel;<br>static int carCount;<br>public Car() { fuel = 0; }<br>public Car(int g) { fuel = g; }<br>public void addFuel() { fuel++; }<br>public void display() { System.out.print(fuel + " "); }<br>}<br>class RaceCar extends Car<br>{ public RaceCar(int g) {carCount++; super(2*g); }<br>}<br>class Demo{<br>public static void main(String[] args){<br>Car car = new Car(5);<br>Car fastCar = new RaceCar(5);<br>car.display();<br>car.addFuel();<br>car.display();<br>fastCar.display();<br>fastCar.addFuel();<br>fastCar.display();<br>System.out.println(RaceCar.carCount); }} | 6M |
| 3 c | Write whether the following statements are true/false, justify your answer<br>i. Abstract class can be instantiated.<br>ii. Abstract class can have more than one abstract method<br>iii. Abstract class enforces the polymorphism<br>iv. If your answer for the question number iii is yes, then write a code to justify your answer.<br>v. The sub class of an abstract class has to implement all the abstract methods of the abstract class, then it can be instantiated?, otherwise, how would you define the sub class? | 10M |

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## ISA EXAM - II

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| Course : **Object Oriented Programming** | USN : |
| Course Code : **25ECSC204/24ECAC207** | Semester : 4 |
| Date of Exam : 26/05/2026 | Duration : 75 mins |
| Marks: 40 | |
| **Note: Answer any two full questions. Each full question carries 20 marks** | |

| Q.No. | Questions | Marks |
|---|---|---|
| Q1.a | Explain the exception handling mechanism of java. | 4 |
| Q1.b | Write the output of the following code snippet and comment on the output<br>`interface Printable {`<br>`  void print();}`<br>`interface Scannable {`<br>`  void scan();}`<br>`class PrinterMachine implements Printable, Scannable {`<br>`  public void print() {`<br>`    System.out.println("Printing document"); }`<br>`  public void scan() {`<br>`    System.out.println("Scanning document");  } }`<br>`public class InterfaceDemo {`<br>`  public static void main(String[] args) {`<br>`    Printable device = new PrinterMachine();`<br>`    device.print();`<br>`    device.scan();   } }` | 6 |
| Q1.c | Each customer owns a wallet. A customer can add money to the wallet and spend money from it. Sometimes, the customer can also send money to another customer's wallet. Write a Java program to create a few customers along with their wallets, then perform the following operations:<br>i. Add money to wallet<br>ii. Spend money from wallet<br>iii. Transfer money between wallets<br>Handle an exception while transferring money such that the transfer amount should not exceed Rs 25000. | 10 |
| Q2.a | Explain the methods put(), get(), remove(), and containsKey() in Java Map interface with examples. | 4 |
| Q2.b | Define a Lambda Expression in Java. Explain its syntax with suitable examples. | 6 |

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| Q2.c | Write the following methods that *return a lambda expression* performing a specified action:<br>A. PerformOperationisOdd(): The lambda expression must return true if a number is odd or false if it is even.<br>B. PerformOperationisPrime(): The lambda expression must return true if a number is prime or false if it is composite.<br>C. PerformOperationisPalindrome(): The lambda expression must return true if a number is a palindrome or false if it is not. | 10 |
| Q3.a | List four method of stream interface and explain their usage in java programs. | 4 |
| Q3.b | Predict the output of the following code: (code below) | 6 |

```
import java.util.*;
import java.util.stream.*;
class StreamNumbersDemo {
  public static void main(String[] args) {
    ArrayList<Integer> marks = new ArrayList<>();
    marks.add(42);   marks.add(17);   marks.add(91);   marks.add(28);
    marks.add(65);   marks.add(10);
    Stream<Integer> markStream = marks.stream();
    Optional<Integer> maxVal = markStream.max(Integer::compare);
    if(maxVal.isPresent())
      System.out.println("Maximum value: " + maxVal.get());
    Stream<Integer> sortedMarks = marks.stream().sorted();
     System.out.print("Sorted values: ");
    sortedMarks.forEach(n -> System.out.print(n + " "));
     System.out.println();
    long count = marks.stream().count();
      System.out.println("Total elements: " + count);
      System.out.print("Even numbers: ");
    marks.stream()
        .filter(n -> n % 2 == 0)
        .forEach(n -> System.out.print(n + " "));
    System.out.println();
        int sum = marks.stream()
              .mapToInt(Integer::intValue)
              .sum();
    System.out.println("Sum of elements: " + sum);   } }
```

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| Q3.c | Write a Java program to perform the following:<br>i) Create a class Employee(empId, empName, department, salary)<br>ii) Create a List object using LinkedList and add a few employee objects to it.<br>iii) Call suitable methods on the LinkedList to obtain:<br>A. First employee added B. Last employee added<br>iv) Call methods to remove:<br>C. First employee added D. Last employee added<br>v) Display all remaining employee details in the list. | 10 |